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# Matrices

## Matrix

$$A = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & \cdots & \cdots & \cdots \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{bmatrix}_{m \times n}$$

Vector addition and scalar multiplication can be done on matrices.

$$\begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{bmatrix} + \begin{bmatrix} b_{11} & b_{12} & \cdots & b_{1n} \\ \vdots & \vdots & \ddots & \vdots \\ b_{m1} & b_{m2} & \cdots & b_{mn} \end{bmatrix} = \begin{bmatrix} (a_{11}+b_{11}) & (a_{12}+b_{12}) & \cdots & (a_{1n}+b_{1n}) \\ \vdots & \vdots & \ddots & \vdots \\ (a_{m1}+b_{m1}) & (a_{m2}+b_{m2}) & \cdots & (a_{mn}+b_{mn}) \end{bmatrix}$$

$$\lambda \cdot \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{bmatrix} = \begin{bmatrix} \lambda a_{11} & \lambda a_{12} & \cdots & \lambda a_{1n} \\ \vdots & \vdots & \ddots & \vdots \\ \lambda a_{m1} & \lambda a_{m2} & \cdots & \lambda a_{mn} \end{bmatrix}$$

Scalar multiplication shrinks or stretches the matrices.

## Matrix multiplication

$$A = \begin{bmatrix} 1 & 2 & 1 \\ 2 & 2 & 3 \end{bmatrix}_{2\times3} \qquad B = \begin{bmatrix} 3 & 6 \\ 4 & 7 \\ 5 & 8 \end{bmatrix}_{3\times2}$$

($b_1$, $b_2$)

$$A \cdot B = \begin{bmatrix} Ab_1 & Ab_2 \end{bmatrix} = \left[ 3\begin{bmatrix}1\\2\end{bmatrix} + 4\begin{bmatrix}2\\2\end{bmatrix} + 5\begin{bmatrix}1\\3\end{bmatrix} \quad 6\begin{bmatrix}1\\2\end{bmatrix} + 7\begin{bmatrix}2\\2\end{bmatrix} + 8\begin{bmatrix}1\\3\end{bmatrix} \right]$$

$$= \begin{bmatrix} 1\cdot3+2\cdot4+1\cdot5 & 1\cdot6+2\cdot7+1\cdot8 \\ 2\cdot3+2\cdot4+3\cdot5 & 2\cdot6+2\cdot7+3\cdot8 \end{bmatrix} = \begin{bmatrix} 16 & 28 \\ 26 & 44 \end{bmatrix}_{2\times2}$$

$$A_{m\times n} \times B_{n\times p} = C_{m\times p}$$

## Laws of matrices

### Addition:

1. $A+B = B+A$ ··· commutativity
2. $c(A+B) = cA + cB$ ··· distributive
3. $A+(B+C) = (A+B)+C$ ··· associativity

### Multiplication:

1. $C(A+B) = CA + CB$ ··· distributive from left
2. $(A+B)C = AC + BC$ ··· distributive from right
3. $A(BC) = (AB)C$ ··· associativity

could be equal in some cases, but not necessarily.

NOTE: $AB \neq BA$ [The vector space changes]

## Laws of exponents over square matrices:

1. $A^n = A\,A \cdots A$ ($n$ times)
2. $A^n A^q = A^{n+q}$
3. $(A^n)^q = A^{nq}$

# Linear maps

A linear map from V to W is a function T: V→W with the following properties:
i. Additivity: T(u+v) = Tu + Tv ∀ u, v ∈ V.
ii. Homogenity: T(λv) = λT(v) ∀ λ ∈ F, v ∈ V.

$T: V \to W$

$T(u+v) = T(u) + T(v)$

$T(\lambda v) = \lambda T(v)$

## Notation

- Set of linear maps from V to W: $L(V, W)$
- Set of linear maps from V to V: $L(V)$

NOTE: $L(V, W)$ is a functional vector space.

Examples:

1. Identity operator: let $I \in L(V)$ s.t. $Iv = v \quad \forall\ v \in V$

$$I(u+v) = u+v = I(u) + I(v)$$
$$I(\lambda v) = \lambda v = \lambda \cdot Iv$$

2. let $T \in L(\mathbb{R}^3, \mathbb{R}^2)$ s.t.

$$T(x, y, z) = (2x - y + 3z,\ 7x + 5y - 6z)$$

3. In general from $\mathbb{F}^n$ to $\mathbb{F}^m$ for $n, m > 0$, let $A_{j,k} \in \mathbb{F}$ for each $j = 1, \dots, m$ and $k = 1, \dots, n$

and $T \in L(\mathbb{F}^n, \mathbb{F}^m)$ s.t.

$$T(x) = T(x_1, x_2, \dots, x_n)$$
$$= (A_{11}x_1 + \dots + A_{1n}x_n,\ \dots,\ A_{m1}x_1 + \dots + A_{mn}x_n)$$

$$\begin{bmatrix} A_{11} & A_{12} & \cdots & A_{1n} \\ A_{21} & A_{22} & \cdots & A_{2n} \\ \vdots & & & \\ A_{m1} & A_{m2} & \cdots & A_{mn} \end{bmatrix}_{m \times n} \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}_{n \times 1} = \begin{bmatrix} A_{11}x_1 + \dots + A_{1n}x_n \\ \vdots \\ A_{m1}x_1 + \dots + A_{mn}x_n \end{bmatrix}_{m \times 1}$$

Matrices are linear maps.

$$AX = T(x)$$

Any linear map $\mathbb{R}^m \to \mathbb{R}^n$ can be represented in matrix form as above.

$T(0) = 0$ for every linear map $T$.

Proof: $T$ is a linear map so it follows additivity.

$$T(0+0) = T(0) + T(0)$$
$$\Rightarrow T0 = T0 + T0$$
$$\Rightarrow T0 = 0$$

# Null spaces

For $T \in L(V, W)$,

$\text{null } T = \{v \in V : Tv = 0\}$

The null space of $T: V \to W$ is the subset of V of those vectors that are mapped to 0 by T.

E.g. If T is zero mapped from $V \to W$, i.e. $Tv = 0 \quad \forall \ v \in V$,

$\text{null } T = V \rightarrow$ subspace of V.

2. $T \in L(\mathbb{R}^3, \mathbb{R}^2)$ s.t. $T(x, y, z) = (2x - y + 3z, 7x + 5y - 6z)$

$$\text{null } T = \left\{ x, y, z \in \mathbb{R}^3 : \begin{matrix} 2x - y + 3z = 0 \\ 7x + 5y - 6z = 0 \end{matrix} \right\}$$

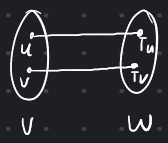
↳ subspace of V.

null T is a subspace of V (domain of T).

## Injectivity (one-to-one)

A function $T: V \to W$ is injective if

$Tu = Tv$ implies $u = v$.



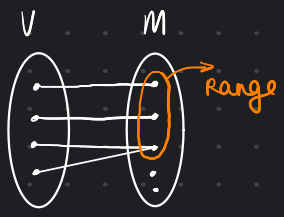

Let $T \in L(V, W)$. Then T is injective iff null $T = \{0\}$.

↳ columns of T are LI.

## Range of T

For $T \in L(V, W)$, the range of T is the subset of W (→ subspace of W) containing those vectors that are equal to Tv for some $v \in V$.

$$\text{range } T = \{Tv : v \in V\}$$




Range = All vectors in W that can be written as Tv for some $v \in V$.

$$\begin{bmatrix} | & & | \\ t_1 & \cdots & t_n \\ | & & | \end{bmatrix} \begin{bmatrix} v_1 \\ \vdots \\ v_n \end{bmatrix}$$

↳ matrix of T ↳ some $v \in V$.

$Tv \ \forall \ v \in V$ = range(T)

If we consider all the values of v possible, we get the span of T.

## Surjectivity (onto)

A function $T: V \to W$ is surjective if

range T = W.

